MODULE 15 · SECTION 2 - INGESTING AND PROCESSING THE DATA

Spark and Hadoop on Google Cloud

Managed Service for Apache Spark: clusters or serverless, Cloud Storage instead of HDFS, workers, autoscaling and metastores.

SECTION 2 of 5 EXAM WEIGHT 25% MODULE 15 of 45

admin@sthithapragna.com

Stop treating a cluster as a building you own. Treat it as **a taxi you call for one journey**.

On-premises Hadoop, or the Google way?

ON-PREMISES

One big cluster, always on

- Data lives in HDFS on the cluster
- Sized for the busiest night
- Shared by every team
- Idle most of the day, still paid for

ON GOOGLE CLOUD

Storage apart, compute on demand

- Data lives in Cloud Storage
- Clusters sized for each job
- One cluster per job or team
- Deleted when the work is done

THE POINT

Separating storage from compute is what makes short-lived clusters possible.

Dataproc has a new name

- Since April 2026 it is **Managed Service for Apache Spark**.
- It covers the old **Dataproc** clusters and **Serverless for Apache Spark**.
- The **CLI still says gcloud dataproc**, and APIs and roles did not change.
- Clusters start, scale and shut down in **about 90 seconds** on average.

> REMEMBER
>
> The exam guide says Dataproc. Answer with the name the question uses.

Serverless or cluster?

	SERVERLESS	CLUSTER
What you manage	Nothing but the job	Machines, software, sizing
Startup	About 50 seconds	About 120 seconds
You pay for	Job run time only	Cluster up time
Runs	Spark batches and sessions	Spark, Hive, Flink, Trino, Kafka
Control	Custom containers	Machine types, SSH, init actions

THE TIEBREAKER

Spark only, intermittent, no ops team: serverless. Hive or deep control: a cluster.

Cloud Storage instead of HDFS

- Jobs read **gs://** paths directly. The connector is **preinstalled**.
- Data **survives the cluster**, so clusters can be deleted freely.
- Spark, Hive, BigQuery and Dataflow all read the **same files**.
- HDFS ties data to machines. Keep it only for **scratch space**, if at all.

> THE TRAP
>
> Lift-and-shift answers that keep data in on-cluster HDFS are usually the trap.

Call a cluster, use it, delete it

- **Job-scoped clusters**: create, run, delete. Sized for that one job.
- **Scheduled deletion**: after 5 minutes to 14 days idle, at a set time, or at a maximum age.
- Orchestrate with **workflow templates** or **Managed Airflow**.
- Idle cluster? **Delete it**. Recreating is as fast as resizing.

> THE SIGNAL
>
> 'Cluster idle most of the day' points to job-scoped clusters or serverless, not autoscaling down.

Primary or secondary workers?

	PRIMARY WORKERS	SECONDARY WORKERS
Role	Compute, plus HDFS DataNodes	Compute only, no HDFS
Type	Standard VMs	Preemptible by default, or Spot
Cost	Full price	Much cheaper
Risk	Stable	Can be reclaimed at any time
Guidance	The steady core	Keep preemptible under half of all workers

REMEMBER

Spot VMs have no 24-hour limit. Expect transient task failures, and raise retries.

Autoscaling policies, and when not to use them

- Scales **workers only**, never masters, from **YARN pending memory**.
- Set the **graceful decommission timeout** longer than the longest job.
- Not for **on-cluster HDFS**, not for **Spark Structured Streaming**, not for idle clusters.
- **Enhanced Flexibility Mode**: scale only secondary workers, decommission timeout 0s.

> THE TRAP
>
> Streaming Spark plus autoscaling is a trap. Google says they are not compatible.

Init actions and a shared metastore

- **Initialization actions**: scripts from Cloud Storage, run as root on every node.
- They run at creation, with a **10-minute** default timeout. Copy public samples to your own bucket.
- **Custom images** bake the software in, instead of installing it each time.
- Short-lived clusters need a **shared metastore**, or table definitions vanish with them.

THE OPTIONS

Dataproc Metastore is a managed Hive metastore. The Lakehouse runtime catalog also offers a Hive endpoint.

Larkspur's nightly settlement cluster

```
gcloud dataproc clusters create lk-settle-nightly \
    --region=europe-west2 \
    --num-workers=4 --num-secondary-workers=3 \
    --secondary-worker-type=spot \
    --autoscaling-policy=lk-secondary-only \
    --dataproc-metastore=projects/lk-data-prod/locations/europe-west2/services/lk-hms \
    --max-idle=30m

# A small, irregular job: no cluster at all
gcloud dataproc batches submit pyspark gs://lk-jobs-prod/tariff_check.py \
    --region=europe-west2
```

3-4 Spot secondary workers, kept under half the total.

6-7 Tables live in a shared metastore. The cluster deletes itself when idle.

REMEMBER

The command line still says dataproc, even though the product has a new name.

Spark and BigQuery together

- The **BigQuery connector** reads tables through the **Storage Read API**.
- Writes are staged in a **Cloud Storage bucket** during the job.
- Serverless Spark gets BigQuery read costs **included**.
- Spark also runs inside **BigQuery Studio notebooks**, on serverless sessions.

> THE SIGNAL
>
> Spark should read BigQuery through the connector, not by exporting tables to files first.

Lift and shift, then modernise

01
Move the data
History into Cloud Storage, compacted. Module 07.

02
Lift and shift
The same Spark and Hive jobs, on clusters. Few code changes.

03
Make it ephemeral
Job-scoped clusters, shared metastore, Spot workers.

04
Go serverless
Small and irregular Spark jobs as batches.

05
Modernise
Hive SQL into BigQuery, tables into Iceberg.

THE PATTERN

Lift and shift first to meet the deadline. Modernise once nothing is at risk.

Phrases that pick the answer

THE QUESTION SAYS	REACH FOR
Idle most of the day	Job-scoped clusters, or serverless
Data must survive the cluster	Cloud Storage, not HDFS
Cut cost on batch workers	Spot secondary workers, under half
Table definitions lost between clusters	A shared metastore
Spark only, no ops team, irregular	Serverless batches
Hive, Trino or Flink alongside Spark	A cluster

THE PATTERN

Data in Cloud Storage, metadata in a metastore, compute only while it is needed.

> After the lift and shift, Larkspur runs a 40-node cluster around the clock. It is busy for about six hours each night and nearly idle the rest of the day.
>
> All data and table definitions already live outside the cluster.

How should the team cut the cost?

A Keep the 40-node cluster running, and buy committed use discounts

B Autoscale the always-on cluster down to two workers in the daytime

C Move the data into HDFS on persistent disks for faster nightly runs

D Create a cluster for each nightly run, and delete it afterwards

◆ PAUSE AND DECIDE

How should the team cut the cost?

A **Committed use discounts**

A discount on eighteen idle hours is still eighteen idle hours.

B **Autoscale down by day**

Google advises deleting idle clusters, not shrinking them.

C **Data into HDFS**

Ties data to machines, which stops the cluster being deleted.

D **A cluster per run**

Pay for six busy hours, and nothing for the idle day.

THE SIGNAL

Idle most of the day, with data and metadata outside: job-scoped clusters.

> Larkspur now creates a fresh cluster for each settlement job. Each new cluster starts with an empty Hive metastore.
>
> Jobs fail because the tables they expect are missing, and analysts cannot find tables created by earlier runs.

What should the team do?

A. Back up each cluster's metastore database, and restore it on the next

B. Attach every cluster to one shared Dataproc Metastore service

C. Add a script to every job that recreates all of the tables first

D. Keep one small permanent cluster that exists only to host the metastore

◆ PAUSE AND DECIDE

What should the team do?

A **Back up and restore**
Fragile, slow, and two overlapping clusters would clash.

B **A shared metastore**
Every cluster sees the same tables, and they outlive each one.

C **Recreate tables per job**
Hundreds of definitions duplicated in every job, and drift.

D **A permanent cluster**
An always-on server just for metadata, which a managed service replaces.

THE SIGNAL

Short-lived clusters losing table definitions need a shared, managed metastore.

> To save money, a Larkspur cluster now runs 2 primary workers and 10 Spot secondary workers.
>
> Jobs have become slow and unpredictable, with many failed tasks whenever Spot capacity is reclaimed.

What should the team change?

A. Keep Spot workers under half of the total, and turn on EFM

B. Make every worker Spot, so that the whole cluster is cheaper

C. Store shuffle data in HDFS on the secondary workers instead

D. Raise the graceful decommission timeout to a full 24 hours

◆ PAUSE AND DECIDE

What should the team change?

A Under half, plus EFM

Google's guidance: a stable core, and a mode built for preemption.

B All Spot

More reclaimed machines, more failures, and no stable core.

C HDFS on secondaries

Secondary workers store no HDFS data at all.

D A 24-hour timeout

Decommission timeouts do not stop Spot capacity being reclaimed.

THE SIGNAL

More than half preemptible, with failures on reclaim: rebalance, and use Enhanced Flexibility Mode.

> Larkspur's data scientists have written a new PySpark job that scores tariff changes. It runs a few times a week, at unpredictable times.
>
> It uses only Spark, and the team has nobody to manage infrastructure.

Where should the job run?

A. A long-running shared cluster, with an autoscaling policy attached

B. A Spark cluster on GKE, run by the central platform team

C. A serverless batch workload, submitted whenever it is needed

D. Rewrite the job in Apache Beam, and run it on Dataflow

◆ PAUSE AND DECIDE

Where should the job run?

A A shared cluster

Always on for a job that runs a few times a week.

B Spark on GKE

Infrastructure for another team, when nobody has time for it.

C A serverless batch

Spark only, irregular, no infrastructure: pay just for each run.

D Rewrite in Beam

A rewrite of working Spark code for no gain.

THE SIGNAL

New, Spark only, irregular, no ops team: serverless batches.

Module 15 in five lines

- Dataproc is now **Managed Service for Apache Spark**. The CLI still says dataproc.
- Data in **Cloud Storage**, not HDFS, so clusters can be **short-lived**.
- **Serverless** for Spark-only, irregular work. **Clusters** for Hive, control, migrations.
- **Spot secondary workers** under half. No autoscaling for HDFS or Structured Streaming.
- Ephemeral clusters need a **shared metastore**. Delete idle clusters.

COMING UP

Next: module 16, getting data into BigQuery, including CDC.